

Introduction to IT and Business Systems

1. Introduction to IT and Business Systems

A. Fundamentals of Information Technology (IT)

- Information Technology (IT) is the use of computers, software, networks, and systems to manage information.
- It involves storing, processing, transmitting, and retrieving data.
- Main components: Hardware, Software, Networks, Data, and People.
- IT is used in banking, education, healthcare, government, and business organizations.

B. Role of IT in Business Operations

- IT automates business tasks such as payroll, billing, accounting, and inventory management.
- Improves communication using email, video conferencing, and enterprise collaboration tools.
- Supports decision-making through data analysis, dashboards, and reporting systems.
- Enhances customer service through online portals and mobile applications.

C. Overview of Business Systems (ERP & CRM)

- ERP (Enterprise Resource Planning) integrates all business functions into one system.
- ERP modules include Finance, HR, Supply Chain, Manufacturing, and Procurement.
- CRM (Customer Relationship Management) focuses on managing customer data and interactions.
- CRM systems help in marketing campaigns, sales tracking, and customer support.

D. Interplay between IT and Business Strategy

- IT helps organizations achieve long-term business goals and digital transformation.
- Aligns technology investments with business objectives.
- Improves efficiency, reduces costs, and increases competitiveness.
- Examples: E-commerce platforms, AI-driven marketing, cloud computing strategies.

2. Data Types, Sources, and Lifecycle

A. Understanding Different Data Types

- Structured Data: Organized in rows and columns, stored in databases and spreadsheets.
- Unstructured Data: Images, videos, emails, and social media posts without fixed format.
- Semi-structured Data: JSON, XML, and log files with partial structure.

B. Common Data Sources

- Internal Data: Sales records, HR data, finance data generated within an organization.
- External Data: Market research, government statistics, social media data.

- Databases: SQL and NoSQL databases used to store large datasets.
- APIs: Provide real-time data from external platforms.

C. Data Collection Methods

- Surveys and questionnaires to collect opinions and feedback.
- Interviews and focus groups for qualitative data.
- Sensors and IoT devices for real-time data collection.
- Transaction systems and web scraping for automated data gathering.

D. Data Lifecycle Management

- Creation: Data is generated or collected from different sources.
- Storage: Data is stored in databases, cloud storage, or data warehouses.
- Usage: Data is processed and analyzed for business insights.
- Archiving: Old data is stored securely for compliance and reference.
- Deletion: Unnecessary or outdated data is securely removed.

E. Data Quality and Integrity

- Accuracy: Data should be correct and error-free.
- Completeness: No missing values or gaps.
- Consistency: Same data across different systems.
- Timeliness: Data should be updated regularly.
- Validity: Data must follow defined rules and formats.

3. Introduction to Business Analyst Role

A. Definition and Responsibilities

- A Business Analyst (BA) bridges the gap between business needs and IT solutions.
- Identifies business problems and proposes solutions.
- Gathers and documents requirements from stakeholders.
- Helps in process improvement and system design.

B. Key Skills for Business Analysts

- Strong communication and presentation skills.
- Analytical and critical thinking ability.
- Problem-solving and decision-making skills.
- Technical skills such as Excel, SQL, Power BI, and Tableau.

C. Business Analysis Methodologies

- Waterfall: Sequential model with fixed phases.
- Agile: Iterative and flexible approach.
- Scrum and Kanban are popular Agile frameworks.

D. Stakeholder Management

- Stakeholders include customers, managers, developers, and vendors.

- BA communicates requirements and project updates.
- Manages expectations and resolves conflicts.

4. Data vs Information

- Data refers to raw facts and figures without context.
- Information is processed and meaningful data.
- Transformation process: Data → Information → Knowledge → Decision.
- Information helps organizations plan, control, and forecast.

5. Data-Driven Decision Making

A. Principles

- Decisions should be based on data and facts.
- Use analytics tools and dashboards for insights.

B. Benefits

- Improves accuracy and reduces human bias.
- Helps in identifying opportunities and risks.
- Increases efficiency and profitability.

C. Steps in Data-Driven Decision Process

- Define the problem or objective.
- Collect relevant data.
- Clean and analyze the data.
- Interpret results and generate insights.
- Make decisions and monitor outcomes.

D. Challenges and Pitfalls

- Poor data quality and incomplete data.
- Privacy and security concerns.
- Bias in data and algorithms.
- Lack of skilled data professionals.

6. Case Discussions / Real-Life Examples

- Netflix uses data analytics for movie and series recommendations.
- Amazon uses data for pricing, logistics, and customer personalization.
- Banks use analytics for fraud detection and credit scoring.
- Hospitals use data to predict diseases and improve healthcare services.

7. Data Governance and Data Ethics

A. Data Governance

- Defines policies and procedures for data management.
- Assigns roles such as Data Owner, Data Steward, and Data Custodian.

- Ensures data security, quality, and compliance.

B. Principles of Data Ethics

- Protect privacy and confidentiality of personal data.
- Ensure transparency in data collection and usage.
- Maintain accountability for data-driven decisions.

C. Regulatory Compliance

- GDPR (General Data Protection Regulation) in Europe.
- CCPA (California Consumer Privacy Act) in the USA.
- DPDP Act (Digital Personal Data Protection Act) in India.

D. Responsible Data Culture

- Train employees on ethical data usage.
- Implement secure IT systems and ethical AI practices.
- Promote awareness of responsible data handling.